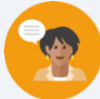


7.8 Using Trigonometric Ratios and the Pythagorean Theorem

Lesson Introduction

 Benchmark	MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Targets	- I can use trigonometric ratios to solve mathematical problems. - I can use the Pythagorean Theorem to solve mathematical problems.
 Benchmark Coherence	Building On: MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.		Working Toward: N/A	
 Essential Questions	- How do I use trigonometric ratios to solve mathematical problems? - How can I use Pythagorean Theorem to solve mathematical problems?			
 Lesson Narrative	Students are focusing on solving mathematical problems involving right triangles. Students will solve problems that involve using a combination of trigonometric ratios and the Pythagorean Theorem.			
 Component Summary	7.8.1 Warm-Up (~5 min.)	Find side lengths of a right triangle (SE p. 416)	7.8.4 Guided Practice (5 min.)	Solve mathematical problems involving right triangle trigonometry and Pythagorean Theorem (SE p. 419)
	7.8.2 Collaboration (5-10 min.)	Find missing angles and sides in overlapping right triangles (SE p. 417)	7.8.5 Your Turn! (10 min.)	Solve mathematical problems involving right triangle trigonometry and Pythagorean Theorem (SE p. 420)
	7.8.3 Guided Instruction (10-15 min.)	Solve mathematical problems involving right triangle trigonometry and the Pythagorean Theorem (SE p. 418)	7.8.6 Wrap-Up (~5 min.)	Students summarize their learning (SE p. 421)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		- Scientific Calculators (Optional) Colored Pencils (Optional) Highlighters	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may incorrectly round when approximating the side length of right triangles using trigonometric ratios. - Students may assume a point is a midpoint in 7.8.5 Your Turn! when solving for the hypotenuse.	- Consider facilitating discussions that allow the students to discuss common misconceptions and how to avoid or overcome these obstacles. - In problems that require several steps, students should not round until the final step. Rounding mid-step could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Group Presentations During 7.8.2 Collaboration, have students display their work for the class. Choose students to spotlight struggles and strategies used when solving for the angle and segment measures.	MA.K12.MTR.1.1: Participate in Effortful Learning The 7.8.4 Guided Practice and 7.8.5 Your Turn! components require students to think about and discover the correct steps in order to find the final answer. Consider using the language from the benchmark clarifications of MA.K12.MTR.1.1 to help students make the connection between making the commitment to learn and having a positive mindset when solving a challenging task.
 Differentiate Instruction	Students may struggle with breaking down the information given in each problem. For auditory learners, consider using question stems during 7.8.2 Collaboration and 7.8.3 Guided Instruction to help students make sense of what the problem is giving them and what the problem is asking of them. For visual learners, consider allowing students to color code given information, such as the side lengths of triangles, while solving mathematical problems.	
Lesson Notes:		

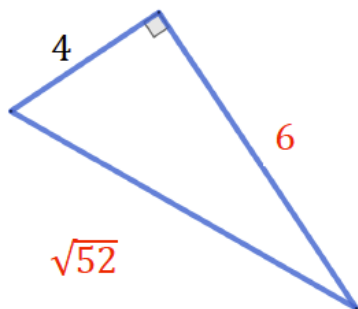
7.8.1 Warm-Up: Critical Thinking

Component Overview: Students are using the digits 0 – 9 to find missing values for the side lengths of a right triangle. Digits used should not be repeated.



7.8.1 Warm-Up: Critical Thinking

1. Use the digits 0-9 one time each to find the missing side lengths of the right triangle. The triangle is not drawn to scale.



Monitor student responses and select a few students with different answers to share how their answer is correct and the strategy used to solve it.

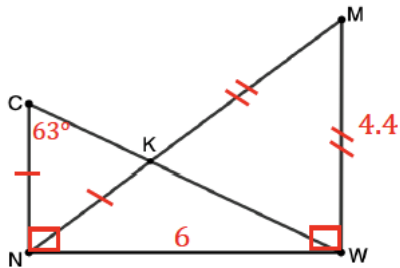
7.8.2 Collaboration: Overlapping Triangles

Component Overview: Students will work in pairs to solve a mathematical problem involving overlapping triangles. Students should use trigonometric ratios and the Pythagorean Theorem to solve for the missing angles and side lengths.

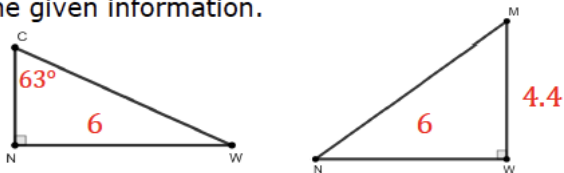


7.8.2 Collaboration: Overlapping Triangles

1. $\triangle MWN$ and $\triangle CNW$ are right triangles that share side $\overline{NW}$ and K is the intersection point of sides $\overline{CW}$ and $\overline{MN}$. In the triangles, the $m\angle MWN = m\angle CNW = 90^\circ$, $MW = 4.4$, $NW = 6$, and $m\angle NCK = 63^\circ$. $\triangle CNK$ is an isosceles triangle with base $\overline{CK}$ and $\triangle MWK$ is an isosceles triangle with base $\overline{WK}$.



- a. Mark the figure with all information given in the problem.
- b. Jasmine redrew the diagram as two separate triangles, as shown. Mark Jasmine's diagram using the given information.



- c. Find the measures of the segments and angles listed in the table. Round to the nearest thousandth. *Answers may vary slightly depending on method and rounding.*

Segment	Length	Angle	Measure
$\overline{MN}$	$MN = 7.44$	$\angle MNW$	36°
$\overline{CN}$	$CN = 3.06$	$\angle WMN$	54°
$\overline{CW}$	$CW = 6.726$	$\angle CWN$	27°
$\overline{KN}$	$KN = 3.06$	$\angle WCN$	63°
$\overline{MK}$	$MK = 4.4$	$\angle CKN$	63°



Allow students to choose their own method of marking given information and congruent parts. Question students about their choice of markings to ensure understanding.

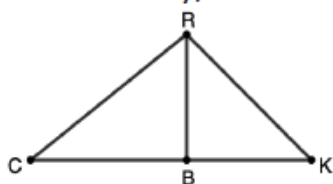
7.8.3 Guided Instruction: Solving Problems With Right Triangles

Component Overview: The teacher will use guided instruction to model solving mathematical problems involving trigonometric ratios and the Pythagorean Theorem.



7.8.3 Guided Instruction: Solving Problems With Right Triangles

1. $\triangle RBK$ and $\triangle RBC$ are right triangles that share side $\overline{RB}$. $\angle RBK$ and $\angle RBC$ are both right angles, $m\angle BCR = 38.7^\circ$, $m\angle RKC = 45^\circ$, and $RB = 8$.
 - a. Use the given information to complete the table. Whenever possible, give lengths as exact values. If necessary, round to the nearest hundredth.



Segment	Length
$\overline{RC}$	12.80
$\overline{CB}$	9.99
$\overline{RK}$	$8\sqrt{2}$
$\overline{KB}$	8.00
$\overline{CK}$	17.99

- b. What is the area of $\triangle CRK$?

$$A = \frac{1}{2}(17.99)(8) = 71.96 \text{ square units}$$

2. Triangle RGH is shown, where

$m\angle RHG = 90^\circ$,
 $m\angle RWH = 50.2^\circ$,
 $m\angle WRG = 16.5^\circ$,
 $m\angle RGH = 33.7^\circ$, and
 $RH = 12$.

What is the approximate perimeter of $\triangle RWG$? Round your answer to the nearest hundredth.

$$RG: \sin(33.7^\circ) = \frac{12}{RG}; \quad RG \approx 21.63$$

$$WG: \tan(33.7^\circ) = \frac{12}{HG}; \quad HG \approx 17.99$$

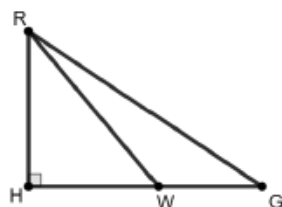
$$\tan(50.2^\circ) = \frac{12}{HW}; \quad HW \approx 9.99$$

$$WG = HG - HW \approx 17.99 - 9.99 \approx 8$$

$$RW: 12^2 + (9.99^2) = RW^2; \quad RW \approx 15.61$$

Therefore, the perimeter of $\triangle RWG$ is:

$$RG + WG + RW \approx 21.63 + 8 + 15.61 \approx 45.24 \text{ units}$$



Common Misconception

Check student answers to ensure that the answers are being rounded correctly. If students are rounding incorrectly, consider facilitating a discussion over how to round.



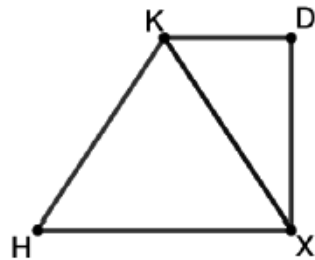
A discussion could provide additional entry points for auditory learners. Consider utilizing sentence stems and provide ample time for students to grapple with academic language:

- I know the altitude of the triangle is _____ because _____.
- I know the base of the triangle is _____.
- Using the area formula for a triangle, I found the area to be _____.

7.8.3 Guided Instruction: Solving Problems With Right Triangles (continued)

3. $\triangle KDX$ and $\triangle HKX$ share side $\overline{KX}$. $\triangle HKX$ is an isosceles triangle with base $\overline{HX}$ and $\triangle KDX$ is a right triangle with $m\angle KDX = 90^\circ$.

If $HX = 13$, $DH = 9$, and $m\angle KXD = 34^\circ$, what is the perimeter of quadrilateral $KDXH$? Round your answer to the nearest hundredth.



First, find KH by finding KX :

$$\cos(34^\circ) = \frac{9}{KX}$$

$$KX = \frac{9}{\cos(34^\circ)} \approx 10.86$$

So, $KH \approx 10.86$.

Next, find KD :

$$\tan(34^\circ) = \frac{KD}{9}$$

$$KD = 9 \cdot \tan(34^\circ)$$

$$KD \approx 6.07$$

$$HK + KD + DX + XH = 10.86 + 6.07 + 9 + 13 = 38.93 \text{ units}$$



For a visual approach to this problem, consider color coding each side using highlighters or colored pencils. This will allow students to track which sides are being solved, as well as which side lengths are congruent. In order to help auditory learners recognize that $\overline{KH}$ and $\overline{KX}$ are congruent, consider using the question stem:

- “I know that _____ and _____ are congruent by the definition of _____.”

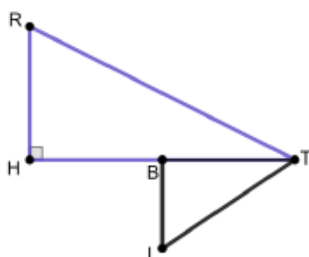
7.8.4 Guided Practice: Right Triangle Trigonometry and the Pythagorean Theorem

Component Overview: The teacher will guide students through the process of solving mathematical problems that involve trigonometric ratios and the Pythagorean Theorem.



7.8.4 Guided Practice: Right Triangle Trigonometry and the Pythagorean Theorem

1. Right triangles RHT and TBL are shown, where B is the midpoint of $\overline{HT}$, $m\angle RHT = m\angle TBL = 90^\circ$, $m\angle RTH = 30^\circ$, $BL = 3$, and $LT = 7$.



- a. What is the length of $\overline{HT}$?

$$4\sqrt{10}$$

- b. What is the area of $\triangle RHT$?

$$\tan(30^\circ) = \frac{RH}{HT}$$

$$\tan(30^\circ) = \frac{RH}{4\sqrt{10}}$$

$$RH = \tan(30^\circ) \cdot 4\sqrt{10} \approx 7.303$$

$$A = \frac{1}{2}(7.303) \cdot 4\sqrt{10}$$

$$A \approx 46.188$$



Allow informal observation throughout the first two activities to determine how “guided” this guided practice should be for you and your classroom. Consider small group instruction for some while others attempt independently, or teacher facilitation of practice with students actively working as a whole group.



This question offers an opportunity to spiral back to special right triangles and the definition of midpoint.

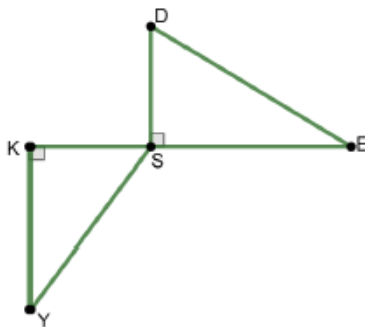
7.8.5 Your Turn!

Component Overview: Students work independently to solve mathematical problems that involve trigonometric ratios and the Pythagorean Theorem.



7.8.5 Your Turn!

1. In the diagram below, triangle DSE and YKS are right triangles that share the vertex S . The $m\angle DSE = m\angle YKS = 90^\circ$, $m\angle DES = 31^\circ$, $m\angle KSY = 53^\circ$, $KE = 37$, and $DS = 10$.



Determine the length of $\overline{KY}$.

$$\tan(31^\circ) = \frac{10}{SE}$$

$$SE = \frac{10}{\tan(31^\circ)}$$

$$SE \approx 16.643$$

$$KS + SE = KE$$

$$KS + 16.643 = 37$$

$$KS \approx 20.357$$

$$\tan(53^\circ) = \frac{KY}{20.357}$$

$$KY = 20.357 \cdot \tan(53^\circ)$$

$$KY \approx 27.015$$



For question 1, students should be instructed to round to the nearest hundredth or write their answer as an exact length. If students are instructed to represent their answer as an exact length, they will not perform any calculations. Students could be asked to show their steps without performing the calculations and then in their final step asked to approximate the length. By doing this, students will demonstrate their understanding in setting up the trig ratios and then show their understanding of how to correctly use their calculator to approximate the solution.

$$\text{Exact length: } \left(\tan 53^\circ \right) \left(37 - \frac{10}{\tan 31^\circ} \right)$$

$$\text{Approximate length: } 27.02$$

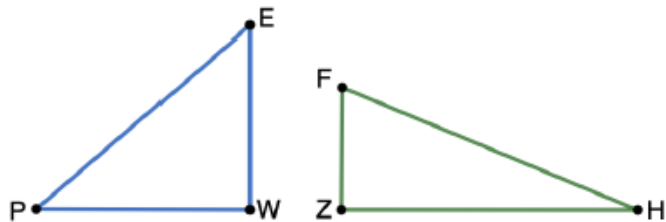


Common Misconception

If students find $KY = 30.74$, then the students assumed that point S was the midpoint of $\overline{KE}$. Consider facilitating a discussion about how information cannot be assumed, unless directly given.

7.8.5: Your Turn! (continued)

2. Triangles PWE and FZH are shown. In $\triangle PWE$, $m\angle WEP = 49.3^\circ$, $PE = 30.6$, and $\angle PWE$ is a right angle.



If $\angle FZH$ is a right angle, FZ is two-thirds the length of EW , and ZH is 1.2 times longer than PW , what is the length of FH ?

$$FZ = \frac{2}{3}EW$$

$$\cos(49.3^\circ) = \frac{EW}{30.6}$$

$$EW = 30.6 \cdot \cos(49.3^\circ)$$

$$EW \approx 19.954$$

$$FZ \approx 13.303$$

$$ZH = 1.2PW$$

$$\sin(49.3^\circ) = \frac{PW}{30.6}$$

$$PW = 30.6 \cdot \sin(49.3^\circ)$$

$$ZH = 1.2(30.6 \cdot \sin(49.3^\circ))$$

$$ZH \approx 27.839$$

$$FH = \sqrt{27.839^2 + 13.303^2}$$

$$FH \approx 30.854$$



For question 2, students should be instructed to round to the nearest hundredth or write their answer as an exact length.



Common Misconception

If students find $PW = 19.95$, then the students used the wrong trigonometric ratio, $\cos 49.3 = \frac{PW}{30.6}$. Have students refer to their notes on the trigonometric ratios and labeling their triangles.

7.8.6 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



7.8.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Answers will vary.